OWASP Foundation

BLT-NetGuardian — Distributed Autonomous Security Scanning Platform

Community-powered security scanning platform with
distributed scanning, real vulnerability detection,
and responsible disclosure workflows

GSoC Project Proposal

Presented to OWASP Foundation
for Google Summer of Code 2026

by

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from

Pune (India)

Google Summer of Code Proposal

About the Contributor

Personal Information

- **Project Title:** Intelligent Vulnerability Validation & Deduplication Engine (BLT-NetGuardian Enhancement)
- **Name of Contributor:** Mohammed Rehan Azim
- **Academic Studies:** B.Tech Computer Science and Engineering (CSE)
- **Year of Graduation:** 2029
- **Enrolled University:** Physics Wallah Institute of Innovation, Pune

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Skill Set

- **Languages:** C++, JavaScript, Java Python, SQL (basic)
- **Frameworks & Technologies:** Web Development, Prompt Engineering, AI-based Systems
- **Tools & Platforms:** GitHub, Docker, Vercel, Google Cloud, Figma
- **Domains:** AI/ML (Applied), Cybersecurity Basics, Product Development, Research Systems
- **Soft Skills:** Analytical Thinking, Team Leadership, Product Design, Public Speaking

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Professional Work Experience

Deaf Link Innovations | Chief Technology Officer (CTO)

Feb 2023 – Present

- Led the development of a **low-cost wearable AI device** for speech-to-vibration conversion, aimed at assisting 18M+ deaf individuals globally.

- Designed real-time communication systems improving accessibility and employability outcomes.
- Managed product architecture, system design, and implementation strategy for scalable deployment.

Firehouse Media | Founder & CEO

Apr 2021 – Jan 2023

- Founded and scaled a freelancing initiative helping **20+ companies grow**, achieving up to **67% increase in sales**.
- Delivered services including web development, UI/UX design, and digital marketing.
- Led cross-functional teams and handled client strategy and execution.

Author – “Master of Web Development”

Apr 2023 – Oct 2024

- Authored and published a web development book with **1,000+ distributions** via NGOs and educational initiatives.
- Focused on simplifying complex technical concepts for beginners and students.

Research & Projects

Nexus Journal | Developer

- Built a **research paper submission platform** enabling students to publish and collaborate on academic work.
- Improved accessibility to research opportunities for high school students.

Honors & Awards

- 🏆 **HackAura 1.0 Hackathon (2026)**
  Secured **2nd Position** by building a trust-based marketplace with fake-review detection and explainable scoring.
- 🏆 **Hewlett Packard Enterprise Hackathon (2022)**
  Achieved **Top 19 rank (All India)** for developing a COVID-19 information platform used by **700+ doctors**.

Additional Information

- **Technical Certifications:** AI Breast Cancer Predictor – The New York Academy of Sciences
- **Languages:** English, Hindi
- **Interests:** AI for Social Impact, Cybersecurity, Open Source Contributions, Research Systems

Links for References

- **GitHub:** https://github.com/MIT-REHAN
- **LinkedIn:** www.linkedin.com/in/rehan-azim-64602a224
- **Projects:** Rehan Azim – Portfolio
- **Email:** rehan5.azim@gmail.com

Project Title

BLT-NetGuardian: Intelligent Vulnerability Validation & Deduplication Engine for Accurate Autonomous Security Scanning

Abstract

BLT-NetGuardian is a distributed, community-powered security scanning platform designed to overcome the limitations of traditional vulnerability scanners, particularly high false-positive rates, redundant findings, and inefficient triage workflows. Existing tools often generate noisy and unreliable results, forcing developers and security teams to spend significant time validating and filtering vulnerabilities.

This proposal introduces an **Intelligent Vulnerability Validation & Deduplication Engine**, a smart validation layer that enhances the accuracy and reliability of scan results. The system combines three key techniques: confidence scoring, automated re-testing, and fingerprint-based deduplication. Each vulnerability is evaluated using rule-based scoring mechanisms, re-tested with multiple payloads to confirm reproducibility, and deduplicated to eliminate redundant findings.

By transforming raw scan outputs into clean, high-confidence vulnerabilities, this system is expected to reduce false positives by 40–70%, significantly decrease manual triage effort, and improve trust in automated security scanning. Integrated within NetGuardian's distributed and Zero-Trust architecture, this solution ensures both scalability and privacy while laying the groundwork for future AI-driven vulnerability analysis and remediation.

Project Details

Organization: OWASP Foundation

Problem Statement

Security scanning tools are essential for identifying vulnerabilities in modern applications; however, they often suffer from a critical limitation, high volumes of false positives and duplicate findings. These inaccuracies significantly increase the manual effort required for validation, overwhelm security teams, and reduce the overall effectiveness of automated scanning systems.

In distributed scanning environments like OWASP NetGuardian, this problem is further amplified due to multiple contributors running scans across diverse environments, leading to redundant reports and inconsistent results. Additionally, the lack of standardized validation mechanisms and prioritization frameworks makes it difficult to distinguish between critical vulnerabilities and low-impact noise.

This results in inefficient triage workflows, delayed remediation, and decreased trust in automated security tools. Therefore, there is a strong need for an intelligent system that can validate, score, deduplicate, and prioritize vulnerabilities in a scalable and privacy-preserving manner while aligning with NetGuardian's zero-trust architecture.

Proposed Solution

My proposed solution is to build an **Intelligent Vulnerability Validation & Deduplication Engine** for BLT-NetGuardian that reduces false positives, removes redundant vulnerabilities, and ensures only high-confidence, actionable findings are surfaced. The system will act as a validation layer between raw scan results and the final dashboard using confidence scoring, automated re-testing, and fingerprint-based deduplication.

Phase 1: System Design & Setup

- Analyze NetGuardian architecture and integrate development environment
- Design validation pipeline and define vulnerability data structures
- Prepare a baseline dataset for testing

Phase 2: Confidence Scoring

- Implement rule-based scoring using detection signals (reflection, execution, response behavior)
- Define scoring rules for common vulnerabilities (XSS, SQLi, etc.)
- Classify vulnerabilities into High, Medium, and Low confidence

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Phase 3: Re-testing & Deduplication

- Build an asynchronous re-testing system with multiple payloads
- Update confidence scores based on re-test results
- Implement fingerprint-based deduplication to remove duplicate findings

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Phase 4: Integration & Finalization

- Integrate all modules into the NetGuardian pipeline
- Perform testing, optimization, and validation
- Prepare documentation and final report

Deliverables

The successful completion of this project will result in the following deliverables:

- A fully functional **Intelligent Vulnerability Validation & Deduplication Engine** integrated into BLT-NetGuardian
- A robust **confidence scoring system** with configurable rules for multiple vulnerability types
- An asynchronous **automated re-testing module** for validating vulnerabilities
- A **fingerprint-based deduplication system** to eliminate redundant findings
- Clean and scalable **API endpoints for dashboard integration**
- Comprehensive **unit and integration tests** ensuring reliability and performance
- Well-structured **documentation and usage guides** for developers and contributors
- A complete **final report, demo, and production-ready pull request** ready for merging

Technology Stack

- **Programming Language:** Python (main backend logic)
- **Backend Framework:** FastAPI (to build APIs and handle requests)
- **Async Processing:** asyncio (for handling re-testing tasks efficiently)
- **Database:** PostgreSQL (to store vulnerabilities and results)
- **Queue System:** Redis (for managing background tasks like re-testing)
- **Static Analysis Tool:** Semgrep (for basic SAST integration)
- **Version Control:** Git & GitHub
- **Containerization:** Docker (for easy setup and deployment)

Code Repository

- **Draft Repository:** Will create and maintain a personal GitHub repository for initial development, testing, and prototypes
- **Final Integration:** Code will be contributed to the **OWASP BLT-NetGuardian** repository

I will follow clean coding practices, proper commit structure, and maintain clear documentation to ensure smooth collaboration and easy integration.

Project Timeline (12 Weeks)

The following table outlines the proposed schedule for the project:

Week	Date Range	Primary Focus / Goal	Deliverable
1	Community Bonding	Understand NetGuardian architecture, discuss design with mentors	Initial setup + design document
2	Phase 1	Setup environment, define data structures, design validation pipeline	Project structure + schema

3	Phase 2	Implement basic confidence scoring logic	Scoring prototype
4	Phase 2	Expand scoring rules for XSS, SQLi, etc.	Scoring module complete
5	Phase 3	Build automated re-testing system (async tasks)	Retesting prototype
6	Phase 3	Improve re-testing + begin deduplication logic	Mid-term report + working retest system
7	Phase 3	Implement fingerprint-based deduplication	Deduplication module
8	Phase 3	Integrate scoring + re-testing + deduplication	Initial integrated pipeline
9	Integration	API development for validated results	API endpoints ready
10	Testing	End-to-end testing, optimization, bug fixing	Optimized system
11	Phase 4	Documentation and usage guides	Documentation draft
12	Final	Final testing, polishing, submission	Final report + demo

Commitment

I am prepared to dedicate **20 hours per week** to this project, treating it as a full-time commitment throughout the GSoC period. I will maintain consistent progress, meet deadlines, and actively communicate updates. I am committed to collaborating closely with mentors, incorporating feedback, and ensuring high-quality contributions to the project.

Contact and Mentorship

I am available for discussions at any time convenient to the mentors and would be happy to walk through my proposal in detail. I look forward to working closely with my assigned mentor(s), actively seeking guidance, and refining the project through continuous feedback and collaboration.